Theory of Computation

More Properties of Regular Languages

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Grammar

- Grammar, $G = (V, \Sigma, P, S)$
  V = Set of Variables
  Σ = Set of terminal symbols
  P = Set of Production Rules
  S = Start Variable
- A regular grammar is any right-linear or left-linear grammar

Regular languages

We have proven:

- Regular languages are closed under:
  - Union
  - Concatenation
  - Star operation
  - Reverse

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- Namely, for regular languages L_1 and L_2

$$\left.\begin{array}{l} \textbf{Union: } L_1 \cup L_2 \\ \textbf{Concatenation: } L_1 L_2 \\ \textbf{Star: } L_1^* \\ \textbf{Reverse: } L_1^R \end{array}\right\} \text{Regular Languages}$$

We will prove:

- Regular languages are closed under:
  - **Complement**
  - **Intersection**

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  - **Intersection**
- Namely, for regular languages L_1 and L_2

$$\left.\begin{array}{ll}\textbf{Complement:} & \overline{L_1} \\ \textbf{Intersection:} & L_1 \cap L_2\end{array}\right\}\text{Regular Languages}$$

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Complement

- **Theorem:** For regular language L the complement $\overline{L}$ is regular
- **Proof:** Take DFA that accepts L and make:
  - nonfinal states final
  - final states nonfinal

  Resulting DFA accepts $\overline{L}$.

Complement

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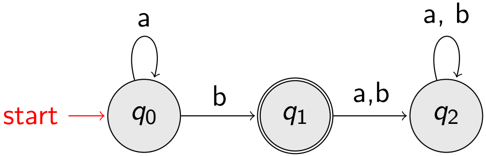
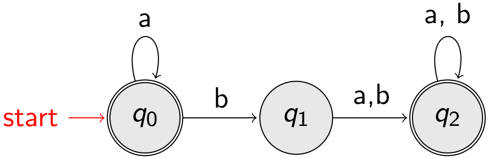
- **Example:**



$L = L(a^* b)$



$\overline{L} =$
$L(a^* + a^* b(a + b)(a + b)^*)$

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  L_1, L_2 Regular

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$$\begin{array}{lll} & L_1,\ L_2 & \text{Regular} \\ \Longrightarrow & \overline{L_1},\ \overline{L_2} & \text{Regular} \end{array}$$

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Standard Representations of Regular Languages

- **Regular Languages**
  - DFAs
  - NFAs
  - Regular Expression
  - Regular Grammar

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Standard Representations of Regular Languages

- **Regular Languages**
  - DFAs
  - NFAs
  - Regular Expression
  - Regular Grammar
- We are given a Regular Language, L
- Language, L is in a standard representation.

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Non-regular languages

- How can we prove that a language is not regular?

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Non-regular languages

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Non-regular languages

- How can we prove that a language is not regular?
- Prove that there is no DFA that accepts L.
- Problem: this is not easy to prove
- Solution: the Pumping Lemma !!!

Non-regular languages

- Given regular language, L and string, w how can we check if $w \in L$?

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Non-regular languages

- Given regular language, L and string, w how can we check if $w \in L$?
- Take the DFA that accepts L and check if w is accepted.

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Non-regular languages

- **Non-regular languages:**
  - $L = \{a^n b^n : n \geq 0\}$
  - $L = \{ww^R : w \in \{a, b\}\}$
  - etc.

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  - L(a*b)
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  - L(b+c(a+b)*)
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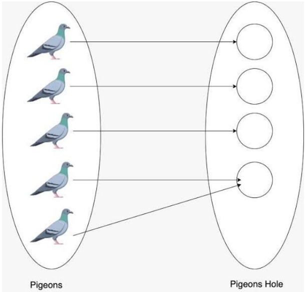
The Pigeonhole Principle

- 5 pigeons and 4 pigeonholes
  A pigeonhole must contain at least two pigeons



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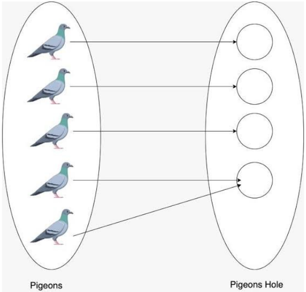
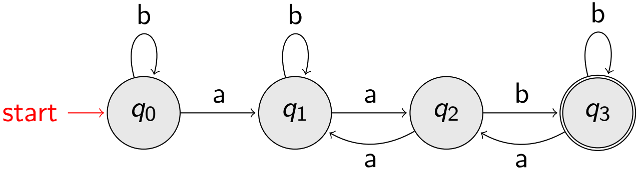


- **Principle:** If there is n number of pigeons and m pigeonholes and $n > m$, There is a pigeonhole with at least 2 pigeons.

The Pigeonhole Principle and DFAs:

- **DFA with 4 states:**



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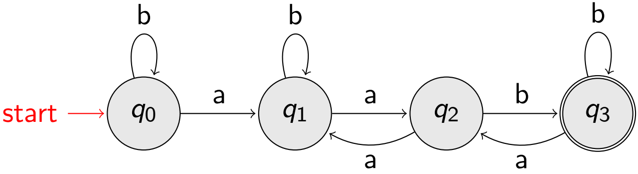

- In walks of strings: $\{a, aa, aab\}$; no state is repeated

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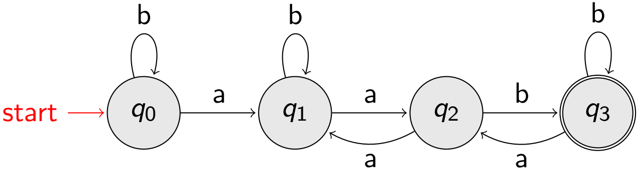
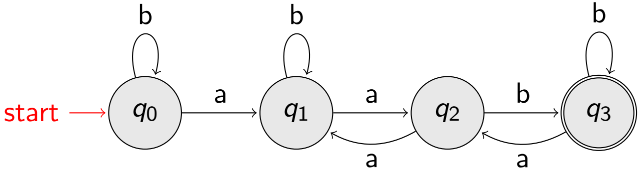


- In walks of strings: $\{a, aa, aab\}$; no state is repeated
- In walks of strings: $\{aabb, bbaa, abbabb, \}$; a state is repeated

The Pigeonhole Principle and DFAs:

- **DFA with 4 states:**



- In walks of strings: $\{a, aa, aab\}$; no state is repeated
- In walks of strings: $\{aabb, bbaa, abbabb,\}$; a state is repeated
- **If string w has length 4:** Then the transitions of string w are more than the states of the DFA. Thus, a state must be repeated.

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- **In general, for any DFA:** String w has length $\geq$ number of states
  A state **q** must be repeated in the walk of **w**.

The Pigeonhole Principle

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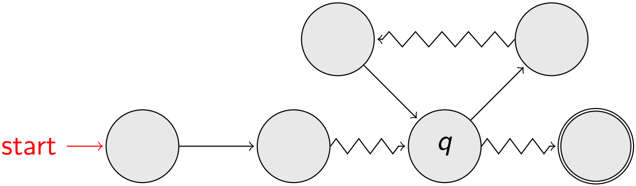
- **walk of w**



- State q will be Repeated.

The Pigeonhole Principle

- In other words for a string w:
  - transitions are pigeons
  - states are pigeonholes

A state **q** must be repeated in the walk of **w**.

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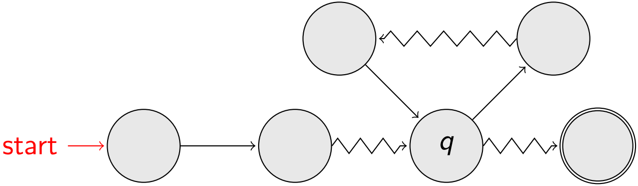

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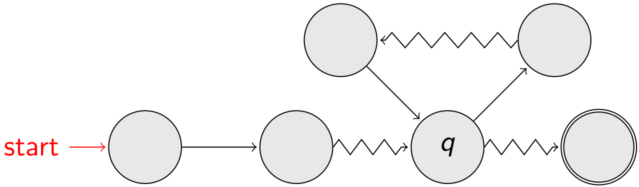
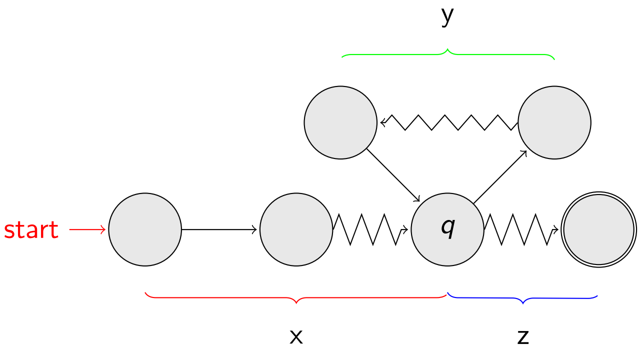
- Take an infinite regular language, L.
  DFA that accepts **L** have **m** number of states.
- Take string w with $w \in L$. There is a walk with label **w**
- If string, **w** has length $|w| \geq m$ number of states of DFA then, from the **pigeonhole principle**:
  a state **q** is repeated in the walk of **w**



The Pumping Lemma

- Let q be the first state repeated; and consider the string $w = xyz$



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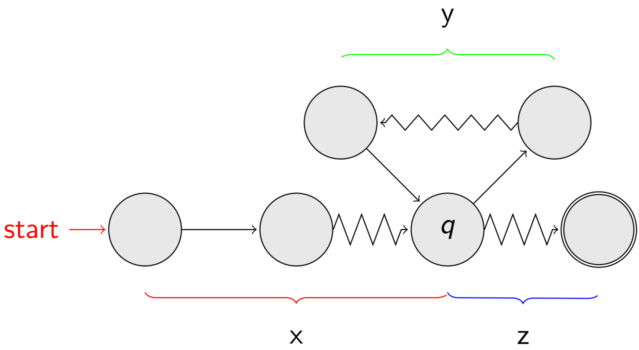

- **Observations:**
  - length, $|xy| \leq m$ number of states
  - length, $|y| \geq 1$
  - The string xz is accepted.

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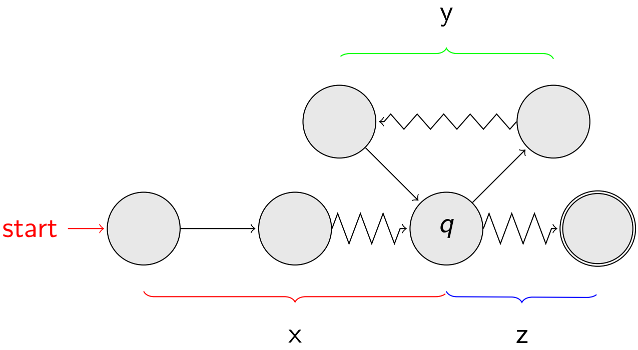

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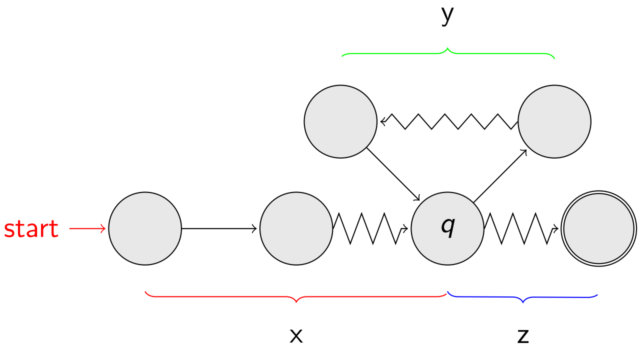

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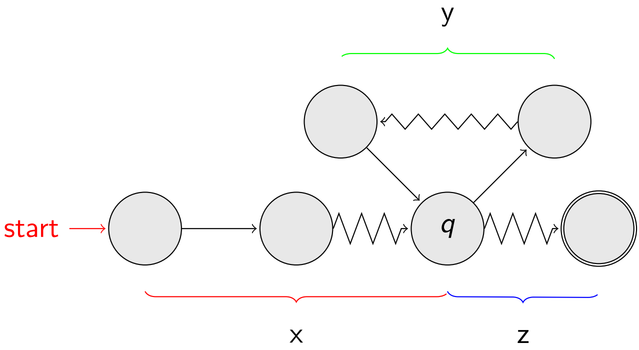

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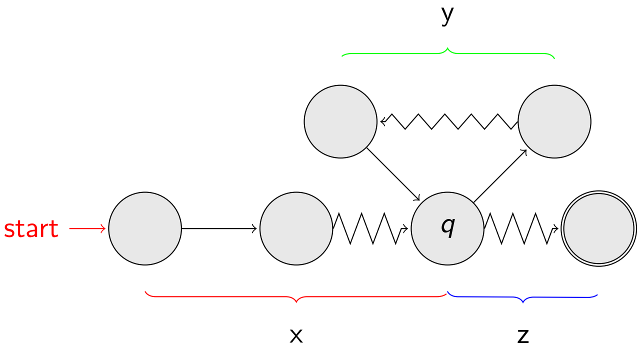

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  - The string $xyyz$ is accepted.

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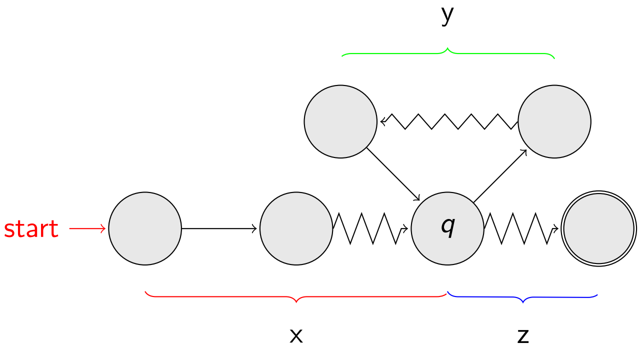

- **Observations:**
  - The string *xz* is accepted.
  - The string *xyz* is accepted.
  - The string *xyyz* is accepted.
  - The string *xyyyz* is accepted.

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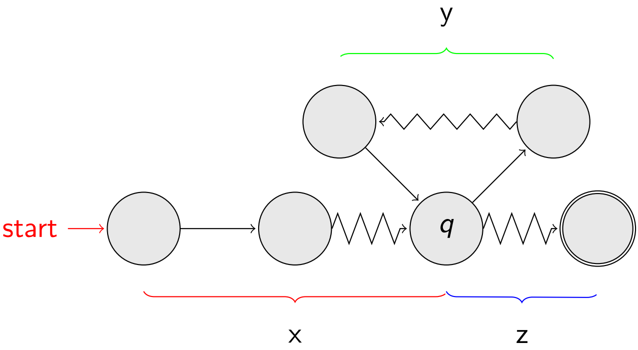
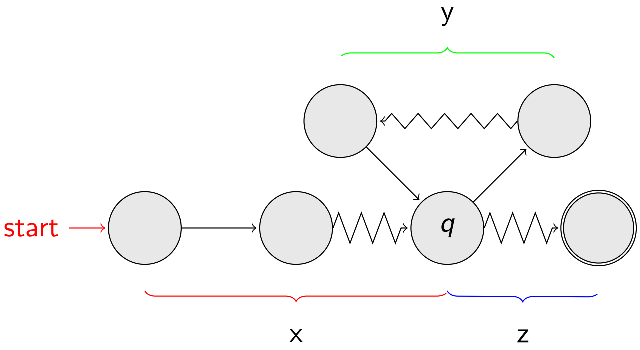


- **Observations:**
  - The string xz is accepted.
  - The string xyz is accepted.
  - The string $xyyz$ is accepted.
  - The string $xyyyz$ is accepted.
  - **In general:** The string xy^iz is accepted; where $i = 0, 1, 2, 3, \ldots$

The Pumping Lemma

- Let q be the first state repeated; and consider the string $w = xyz$



- **Observations:**
  - **In general:** The string xy^iz is accepted; where $i = 0, 1, 2, 3, \ldots$
  - $xy^iz \in L$; $i = 0, 1, 2, 3, \ldots$

The Pumping Lemma:

- Given an infinite regular language, L.
- there exists an integer m,
- for any string $w \in L$ with length $|w| \geq m$
- we can write: $w = xyz$
- with $|xy| \leq m$ and $|y| \geq 1$
- **Such that:** $xy^iz \in L$, $i = 0, 1, 2, 3, \ldots$

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  - Pick a string, w such that: $w \in L$, length $|w| \geq m$
  - we pick $w = a^m b^m$; write w in the form of xyz
  - From the Pumping Lemma $|xy| \leq m$ and $|y| \geq 1$

$$\boxed{xyz = a^m b^m = \overbrace{\underbrace{a \dots a}_{x}\underbrace{a \dots a}_{y}\underbrace{a \dots a}_{}}^{m}\!\!\!\!\!\underbrace{\overbrace{b \dots b}^{m}}_{z}}$$

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$$\boxed{xyz = a^m b^m = \underbrace{a \ldots a}_{x}\underbrace{a \ldots a}_{y}\underbrace{a \ldots ab \ldots b}_{z}}$$

$$\overbrace{a \ldots aa \ldots aa \ldots a}^{m}\overbrace{b \ldots b}^{m}$$

  - Thus, $y = a^k, k \geq 1$
  - From the Pumping Lemma: $\boxed{xy^iz \in L; i = 0, 1, 2, \ldots}$
  - But $xy^iz \notin L$; for i = 2.
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(with $\overbrace{a \ldots aa \ldots aa \ldots a}^{m}\overbrace{b \ldots b}^{m}$)

  - Thus, $y = a^k, k \geq 1$
  - From the Pumping Lemma: $\boxed{xy^iz \in L; i = 0, 1, 2, \ldots}$
  - But $xy^iz \notin L$; for i = 2.
    This is a Contradiction
  - Therefore: Our assumption that L is a regular language is not true.
  - Conclusion: L is not a regular language

- **Example-2:**
  Proof that The language $L = \{ww^R : w \in \Sigma^*\}$ **is not regular.**
  $\Sigma = \{a, b\}$

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- **Example-3:**
  Proof that The language $L = \{a^n b^l c^{n+l} : n, l \geq 0\}$ **is not regular.**

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- **Example-4:**
  Proof that The language $L = \{a^{n!} : n \geq 0\}$ is not regular.

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